

# Appendix to: Modelling of a Galactic Flyby and of Stellar Formation On Stellar Formation

Anvel (Marguerite) de Sainte Marie d'Agneaux

## **1 Introduction**

As mentioned in the report which this is an appendix to, a stellar model must include a more detailed equation of acceleration than a galactic flyby one. In this model, we include gravitational acceleration, the pressure gradient and the viscosity due to the other particles. Without this last one, the contraction and expansion of the star would never stop after the initial collapse. It slows it down until the star can ultimately contract.

In this appendix, we will see how this model was implemented, then some results with different initial parameters for the simulations. Finally, we will take a look at what worked and what could be improved to model stellar formation as accurately as possible. Part of this appendix will be taken or inspired from the initial report for ease of understanding.

## **2 Implementation**

## **3 Results**

## **4 Discussion**

## **5 Conclusion**